

## **Two-week progress update**

Building the Tonic experiment from the DSA lesson

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The simple story: earlier dsa-stdexec experiments showed how to test accelerator overhead by removing one software layer at a time. Over the last two weeks, I made the Rust IDX path clean enough to run that style of experiment inside Tonic.

## The big picture

### What we learned before

1. DSA can be fast when submission cost is amortized.
2. Then the software path becomes visible.
3. The useful method was not guessing; it was comparing controlled layers.

### What I did now

1. Cleaned up the Rust IDXID crates.
2. Built a clearer async memmove path.
3. Proved one DSA operation and one IAX operation on hardware.

### What this enables

1. A Tonic experiment with the same discipline.
2. Ordinary path, software-control path, and IDXID path can be compared.
3. The claim can stay small until the data supports more.

Small claim for today: the Rust IDXID path is cleaner and has a limited real-hardware proof. I am not claiming a Tonic speedup yet.

## Reminder: why the old DSA experiment mattered

### Layer-removal result

| Path         | Throughput | Per-op  | vs base |
|--------------|------------|---------|---------|
| Full stdexec | 26.3 Mpps  | 38.0 ns | 1.00x   |
| Direct path  | 41.6 Mpps  | 24.0 ns | 1.58x   |
| Reusable ops | 59.9 Mpps  | 16.7 ns | 2.28x   |

#### The key number

Removing framework layers cut per-operation cost by about 56%: from 38.0 ns to 16.7 ns. At low concurrency, the reusable path reached 84 Mpps.

### Real hardware context

#### DSA lower bound

A later hardware-floor run reached 48.4 Mops/s for 64 B memmove with pipelined batching.

#### What that means

The device is not the only problem. Once hardware work is cheap enough, software structure can decide whether offload helps.

This is the method I want to reuse: build a ladder of comparable paths, remove one cost at a time, and only then say where the bottleneck is.

## Reusing that method for Tonic

| Experiment step       | What dsa-stdexec did                                                           | Tonic version                                                            |
|-----------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1. Baseline           | Full stdexec path with the normal sender/receiver stack.                       | Ordinary Tonic path with instrumentation off for throughput.             |
| 2. Software controls  | Direct and reusable strategies removed<br>framework costs one layer at a time. | Pooled buffers, copy-minimized paths, and lower-overhead stage counters. |
| 3. Hardware path      | Real DSA run checked whether the lower software cost transfers to hardware.    | Prepared-host IDXD path through the same workload and artifact format.   |
| 4. Matched comparison | Same operation, size, concurrency, and strategy labels.                        | Same payload shape, size, concurrency, runtime, and endpoint split.      |

The next Tonic experiment should be a matched ladder: ordinary Tonic, software control Tonic, then IDXD Tonic.

## What changed in the last two weeks

### Cleaner measurement boundary

1. The current Tonic package now says plainly what it can and cannot prove.
2. The current rows do not show an IDXD win.
3. The workflow is still useful because it is rerunnable and reviewable.

### Cleaner Rust path

1. idxd-rust is now the safe Rust crate.
2. idxd-sys is the raw UAPI/MMIO crate.
3. Async memmove now uses explicit owned buffers.

### Small hardware proof

1. IdxdSession<Dsa> ran DSA memmove.
2. IdxdSession<Iax> ran IAX crc64.
3. A small release-mode benchmark produced positive rows.

I see this as preparation work for the right experiment, not as the final application result.

## Tonic today: useful, but not a speedup claim

The current Tonic result is negative or inconclusive: the package proves the workflow, but the current rows do not show IDX D beating the ordinary path.

### What is already useful

1. Software path validation exists.
2. IDX D verifier gate exists.
3. JSON / CSV / markdown comparison package exists.
4. Current rows show IDX D at roughly 0.003x–0.653x of software throughput.

### What is missing

1. A fresh prepared-host IDX D rerun.
2. Matched ordinary vs IDX D artifacts from the same run context.
3. Lower-overhead stage attribution.
4. A clear crossover rule for when offload helps.

### How I would say this to an advisor

I do not want to sell a speedup. I want to say that the comparison machinery is now honest enough to run the next experiment cleanly.

## Why the Rust cleanup matters for the experiment

### Before

1. Old names like `dsa-ffi` and `idxd-bindings` made ownership unclear.
2. Async request helpers hid too much buffer behavior.
3. Some proof scripts and APIs were hard to explain as one clean path.

### Now

1. `AsyncMemmoveRequest::new(source: Bytes, destination: BytesMut)` makes ownership explicit.
2. Direct Tokio completion owns accepted operations until completion.
3. Errors expose phase, retry, completion, and validation metadata without dumping payload bytes.

This matters because Tonic offload will fail if the memory ownership and completion path are unclear. The cleanup makes the next experiment easier to trust.

# Small hardware proof through the new path

| Evidence        | Target      | Device         | Bytes | Iters | Mean latency | Rate          |
|-----------------|-------------|----------------|-------|-------|--------------|---------------|
| Operation proof | dsa-memmove | /dev/dsa/wq0.0 | 64    | n/a   | completed    | pass          |
| Operation proof | iax-crc64   | /dev/iax/wq1.0 | 64    | n/a   | completed    | crc ok        |
| Small bench     | dsa-memmove | /dev/dsa/wq0.0 | 4096  | 1000  | 6,837 ns     | 146,246 ops/s |
| Small bench     | iax-crc64   | /dev/iax/wq1.0 | 4096  | 1000  | 2,178 ns     | 459,064 ops/s |

Verifier

pass

operation proof and small benchmark both passed

Build

release

measurements were collected in release mode

Failures

0 / 2000

benchmark operations completed without failed rows

This is a proof that the path works for two representative operations. It is not a full performance study.



## The next Tonic experiment I want to run

### Matched workload ladder

1. Start with ordinary Tonic, instrumentation off.
2. Rerun the same points with software controls: pooled buffers and copy-minimized mode.
3. Rerun the same points through the IDX path.
4. Compare only matched size, payload shape, concurrency, runtime, and endpoint setup.

### Evidence to collect

1. Throughput and p99 from instrumentation-off runs.
2. Stage attribution from lower-overhead counters.
3. perf counters or flamegraphs for CPU explanation.
4. Artifact package with ordinary, software-control, and IDX rows side by side.

This mirrors the old DSA experiment: baseline, remove software overhead, add real hardware, then compare matched rows.

## Which Tonic points should be first?

### 4 KiB structured

The pooled-buffer control previously improved throughput by about 2.51x.

Good first test for buffer policy.

### 1 MiB structured

The earlier run was strongly memory-bound: about 63.7% memory-bound in the refined pass.

Good first test for copy movement.

### 64 KiB compression

Compression is payload-sensitive: structured was 0.585x, random collapsed to 0.112x.

Use only as a gated follow-up.

My preferred first pass is 4 KiB and 1 MiB uncompressed. Compression should come after the copy/buffer story is clean.

## My struggle with the agent — why it helped

This was not criticism. The useful loop was: the agent kept the details safe, and I kept steering the story toward what I could explain to my advisor.

### Agent default

1. many artifacts and exact paths
2. careful verification language
3. lots of small true statements
4. too much repo-internal wording

### My revisions

1. make one small claim
2. sound like a human talk
3. connect to the old DSA method
4. explain the next experiment

### Why it worked

1. detail stayed available
2. claims stayed honest
3. the narrative became clearer
4. collaboration improved the slide

The final story came from combining agent memory with human judgment: keep the evidence, but present the research argument.

## What I want feedback on

### Decision question

Does this experiment ladder answer the advisor-level question:

**when does the software path erase the benefit of accelerator offload?**

If yes, I should spend the next step on the prepared-host Tonic rerun.

### Small conclusion

1. The old DSA result gave the method.
2. The last two weeks made the Rust IDX path usable for that method.
3. The next experiment should decide whether the method carries into Tonic.

Bottom line: I am not presenting a Tonic speedup result yet. I am presenting a cleaner path to run the right experiment, with enough hardware proof to make that next experiment credible.